Berry Phase and Non-Abelian Holonomy

From a spin in a rotating field to the tripod atom's U(2) gauge field

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Abstract

Geometric phases are computed here without ever differentiating an eigenvector. For a spin-1/2 in a precessing field the gauge-invariant discrete Berry phase agrees with $-\pi(1-\cos\theta)$ to 2.5×10^{-5} at $M=400$ points, and we derive its finite-M closed form, whose leading error $(\pi^3/3)\sin^2\theta\cos\theta/M^2$ matches the numerics to every printed digit. The lattice Chern number of the sphere is ∓1 to twelve digits. A fourth-order Magnus integrator, checked against the exact rotating-frame solution to 5×10^{-10}, shows that the dynamically extracted phase deviates from the Berry phase linearly in Ω/ω_0 with slope $-(\pi/2)\sin^2\theta$, and that the Aharonov–Anandan phase of the exactly cyclic state is exact at any speed. For the tripod atom, whose two degenerate dark states carry a U(2) gauge field, the polar-decomposed Wilson loop is identical to parallel transport and is confirmed by integrating the full four-level Schrödinger equation ($1-F=6.9\times10^{-7}$, leakage 1.2×10^{-10}). Traversing two loops in opposite order leaves the atom in states with overlap 0.25, while an Abelian control with real couplings commutes to 10^{-13}.

1 Introduction

When a Hamiltonian $H(\mathbf{R})$ is carried slowly around a closed loop in parameter space, an eigenstate returns to itself up to a phase. Besides the dynamical phase $-\int E\,dt$ there is a *geometric* part, the Berry phase [1], which depends only on the path and not on how fast it is traversed. Simon [3] identified it as the holonomy of a U(1) connection, and Wilczek and Zee [2] showed that a d-fold degenerate level carries a U(d) connection whose holonomies are matrices that need not commute. Today this structure underlies band topology [5] and synthetic gauge fields for neutral atoms [7].

Numerically, the obstacle is that an eigensolver returns each eigenvector with an arbitrary phase (or, in a degenerate subspace, an arbitrary basis). Any formula that differentiates eigenvectors is therefore unusable. The goal of this project is to compute Abelian and non-Abelian geometric phases with formulas in which this arbitrariness cancels identically, and to check them against closed forms and against direct time evolution.

Contributions.

1. The discrete Berry phase and its exact finite-M closed form, which gives a machine-precision test at every resolution, plus the lattice Chern number.
2. An exactly unitary fourth-order Magnus integrator, and the non-adiabatic expansion of the extracted phase, with the Aharonov–Anandan phase as its exact generalisation.
3. A polar-decomposed Wilson loop for the tripod dark subspace, the analytic connection and curvature, and a direct four-level simulation that confirms it, including the T^{-2} versus T^{-4} dependence on ramp shape.
4. A demonstration that the order of two loops matters (non-Abelian holonomy), with an Abelian control.

Table 1: Summary of the main results.

Quantity	Computed	Analytic
Berry phase, $M = 400$, five cone angles	max error 2.49×10^{-5}	$-\pi(1 - \cos\theta)$
Discretisation error	matches closed form to 4 digits	$(\pi^3/3)\sin^2\theta\cos\theta/M^2$
Random phase on every u_j	change $\leq 2.2 \times 10^{-15}$	gauge invariant
Reversed loop	$\lvert\gamma_{\text{rev}} + \gamma\rvert \leq 1.3 \times 10^{-15}$	sign flips
Chern number aligned / anti-aligned	-1.000000000000 / $+1.000000000000$	∓ 1
Magnus vs exact, $\Omega/\omega_0 = 0.1$	8.2×10^{-9} (1000 steps), 5.2×10^{-10} (2000)	4th order
Non-adiabatic slope, $\theta = 0.9$	-0.9643	$-(\pi/2)\sin^2\theta = -0.9638$
Aharonov–Anandan, $\Omega/\omega_0 \leq 0.8$	to 3.1×10^{-7}	$-\pi(1 - \cos\theta')$
Tripod $\lVert[A_\varphi, A_S]\rVert$, $\theta = 1.3$	0.351231	$\sqrt{2}\cos\theta\sin^2\theta = 0.351231$
Four-level evolution vs Wilson loop	$1 - F = 6.9 \times 10^{-7}$, leakage 1.2×10^{-10}	$\to 0$
Loop order, overlap of final states	0.2523 (Wilson), 0.2508 (evolution)	< 1
Abelian control	1.000000000000	1

2 Theory and derivations

2.1 Conventions

We set $\hbar = 1$ and write the Schrödinger equation as $i\partial_t\psi = H\psi$. The Berry connection is $A = i\langle u|\nabla u\rangle$, which is real because $\langle u|u\rangle = 1$ implies $\mathrm{Re}\langle u|\nabla u\rangle = 0$. The Berry phase is $\gamma = \oint A \cdot d\mathbf{R}$ and enters the state as $e^{+i\gamma}$. For a degenerate subspace with orthonormal basis $\{D_a\}$, $A^{ab}_\mu = i\langle D_a|\partial_\mu D_b\rangle$ and the holonomy is $U = \mathcal{P}\exp\big(i\oint A\big)$, with later times to the left, acting on the components c of $\psi = \sum_b c_b D_b$.

2.2 Adiabatic theorem and the Berry phase

Let $|n(\mathbf{R})\rangle$ be a non-degenerate eigenstate, $H(\mathbf{R})|n\rangle = E_n(\mathbf{R})|n\rangle$, and let $\mathbf{R}(t)$ vary slowly. Try

$$\psi(t) = c(t)\, e^{-i\int_0^t E_n\, dt'}\, |n(\mathbf{R}(t))\rangle. \tag{1}$$

Substituting into the Schrödinger equation, the term $E_n\psi$ cancels on both sides and what remains is

$$i\dot{c}\,|n\rangle + ic\,\dot{\mathbf{R}}\cdot|\nabla n\rangle = (\text{terms in other eigenstates}). \tag{2}$$

Projecting on $\langle n|$ and neglecting transitions to other levels (the adiabatic approximation),

$$\dot{c} = -c\,\langle n|\nabla n\rangle\cdot\dot{\mathbf{R}} = i\,c\,A\cdot\dot{\mathbf{R}}. \tag{3}$$

Integrating around a closed loop C gives $c(T) = e^{i\gamma}c(0)$ with

$$\gamma = \oint_C A\cdot d\mathbf{R}. \tag{4}$$

The integral contains no time: it is invariant under any reparametrisation of the loop, and so independent of the speed. This is what makes the phase *geometric*.

Gauge transformation. Under $|n\rangle \to e^{i\chi(\mathbf{R})}|n\rangle$ the connection changes as $A \to A - \nabla\chi$, and γ changes by $-\oint\nabla\chi\cdot d\mathbf{R}$. For a single-valued gauge this is a multiple of 2π, so $e^{i\gamma}$ is gauge invariant. On an open path the change is $\chi(\text{end}) - \chi(\text{start})$, which is why only closed loops define a physical phase.

Curvature. By Stokes' theorem $\gamma = \int_S F\, dS$, where $F = \nabla\times A$ is the Berry curvature. F is gauge invariant locally, not just modulo 2π.

2.3 Spin-1/2 in a precessing field

$$H(t) = \frac{\omega_0}{2}\,\hat{n}\cdot\boldsymbol{\sigma}, \qquad \hat{n} = (\sin\theta\cos\varphi,\ \sin\theta\sin\varphi,\ \cos\theta), \qquad \varphi = \Omega t. \tag{5}$$

The aligned eigenstate, in a gauge that is single-valued in φ, is

$$|+\rangle = \begin{pmatrix} \cos\frac{\theta}{2} \\ e^{i\varphi}\sin\frac{\theta}{2} \end{pmatrix}. \tag{6}$$

Only the lower component depends on φ, so

$$\langle +|\partial_\varphi +\rangle = i\sin^2\tfrac{\theta}{2}, \tag{7}$$

$$A_\varphi = i\langle +|\partial_\varphi +\rangle = -\sin^2\tfrac{\theta}{2}. \tag{8}$$

Integrating over $\varphi \in [0, 2\pi)$ at fixed θ,

$$\gamma_+ = -2\pi\sin^2\tfrac{\theta}{2} = -\pi(1-\cos\theta) = -\tfrac{1}{2}\,\Omega_{\text{solid}}, \tag{9}$$

where $\Omega_{\text{solid}} = 2\pi(1-\cos\theta)$ is the solid angle of the cone. The anti-aligned state gives $\gamma_- = -\pi(1+\cos\theta) \equiv +\pi(1-\cos\theta) \pmod{2\pi}$.

Monopole. With $A_\theta = 0$ the curvature is

$$F_{\theta\varphi} = \partial_\theta A_\varphi = -\tfrac{1}{2}\sin\theta. \tag{10}$$

Over the whole sphere $\int F\,d\theta\,d\varphi = -2\pi$. The curvature is that of a magnetic monopole of charge $-\frac{1}{2}$ sitting at the degeneracy $\omega_0 = 0$, and its total flux defines the Chern number

$$C = \frac{1}{2\pi}\int_{S^2} F = -1 \ \ \text{(aligned)}, \qquad +1 \ \ \text{(anti-aligned)}. \tag{11}$$

Two gauges and the Dirac string. The gauge above is singular at the south pole: at $\theta = \pi$ the state is $(0, e^{i\varphi})$, whose phase depends on the direction of approach. The alternative gauge

$$|+\rangle_S = e^{-i\varphi}|+\rangle = \begin{pmatrix} e^{-i\varphi}\cos\frac{\theta}{2} \\ \sin\frac{\theta}{2} \end{pmatrix} \tag{12}$$

is smooth at the south pole but singular at the north pole. It has $A^S_\varphi = A_\varphi + 1 = \cos^2\frac{\theta}{2}$, and its Berry phase $2\pi\cos^2\frac{\theta}{2} = 2\pi + \gamma_+$ agrees with Eq. (9) modulo 2π. No single gauge is smooth on the whole sphere; the two patches differ by the winding factor $e^{-i\varphi}$ on their overlap, and that winding is the Chern number. This is the Dirac-string picture of the monopole.

Higher spin. For a spin j in the state $|m\rangle$ along $\hat{n}$, the same calculation with rotation matrices gives $\gamma_m = -m\,\Omega_{\text{solid}}$. The spin-1/2 result is the case $m = \frac{1}{2}$, and the monopole charge is $-m$.

Adiabatic parameter. Non-adiabatic transitions are driven by the matrix element of the time derivative between the two levels. For the precessing field, $\partial_t = \Omega\,\partial_\varphi$, and with $|-\rangle = (-e^{-i\varphi}\sin\frac{\theta}{2}, \cos\frac{\theta}{2})$,

$$\left|\langle -|\partial_t|+\rangle\right| = \frac{\Omega}{2}\sin\theta. \tag{13}$$

Dividing by the gap ω_0 gives the adiabatic parameter $\frac{1}{2}\varepsilon\sin\theta$ with $\varepsilon = \Omega/\omega_0$. Its square appears in the leakage, and the same $\sin^2\theta$ appears in the first non-adiabatic correction to the phase below.

2.4 Discrete Berry phase and its exact finite-M value

Sample the loop at M points $\mathbf{R}_j$ with eigenvectors u_j. For neighbours,

$$\langle u_j|u_{j+1}\rangle = 1 + \langle u|\nabla u\rangle \cdot \delta\mathbf{R} + O(\delta R^2) = e^{-iA\cdot\delta\mathbf{R}} + O(\delta R^2). \tag{14}$$

Multiplying the links around the loop gives $e^{-i\gamma}$, hence

$$\gamma = -\operatorname{Im}\ln \prod_{j=0}^{M-1} \langle u_j|u_{j+1}\rangle, \qquad u_M \equiv u_0. \tag{15}$$

Each u_j appears once as a bra and once as a ket, so an arbitrary phase on any u_j cancels exactly, at every M.

Exact value on the cone. With $a = \sin^2\frac{\theta}{2}$ and $\delta = 2\pi/M$, every link is the same number,

$$\langle u_j|u_{j+1}\rangle = \cos^2\tfrac{\theta}{2} + \sin^2\tfrac{\theta}{2}\, e^{i\delta} = 1 - a + a\, e^{i\delta}, \tag{16}$$

so the discrete phase is exactly

$$\gamma_M = -M \arg\left(1 - a + a\, e^{i\delta}\right). \tag{17}$$

Leading error. Expanding the argument in δ,

$$\arg(1 - a + ae^{i\delta}) = a\delta - a(1-a)(1-2a)\frac{\delta^3}{6} + O(\delta^5). \tag{18}$$

The first term reproduces $\gamma = -2\pi a$. Using $a(1-a) = \frac{1}{4}\sin^2\theta$ and $1 - 2a = \cos\theta$, the second gives

$$\gamma_M - \gamma = \frac{\pi^3}{3}\,\frac{\sin^2\theta\cos\theta}{M^2} + O(M^{-4}). \tag{19}$$

The error is $O(M^{-2})$ and vanishes at $\theta = \pi/2$, where the δ^3 coefficient changes sign.

2.5 Lattice Chern number

On a (θ, φ) grid define, for each plaquette with corner states $u_1, \ldots, u_4$ taken counter-clockwise, the link variables $U_{ij} = \langle u_i|u_j\rangle/|\langle u_i|u_j\rangle|$ and the flux

$$F_\square = -\arg\left(U_{12}U_{23}U_{34}U_{41}\right) \in (-\pi, \pi]. \tag{20}$$

Each $F_\square$ is gauge invariant, and each link appears in two neighbouring plaquettes with opposite orientation. In the sum over a closed surface every link therefore cancels, except for the multiples of 2π that the principal value of arg discards. Hence $\sum_\square F_\square = 2\pi C$ with C an exact integer [8], provided each plaquette flux is smaller than π.

2.6 Rotating frame: exact solution

The precessing Hamiltonian is a rotation of the static one,

$$H(t) = R(t)H(0)R^\dagger(t), \qquad R(t) = e^{-i\Omega t\sigma_z/2}. \tag{21}$$

Write $\psi = R\chi$. Using $\dot{R} = -\frac{i\Omega}{2}\sigma_z R$ and the fact that R commutes with σ_z,

$$i\dot{\chi} = \Big(H(0) - \frac{\Omega}{2}\sigma_z\Big)\chi \equiv H_{\text{rot}}\,\chi. \tag{22}$$

$H_{\rm rot}$ is time independent, so the exact solution is

$$\psi(t) = e^{-i\Omega t\sigma_z/2}\, e^{-iH_{\rm rot}t}\, \psi(0). \tag{23}$$

With $\varepsilon = \Omega/\omega_0$,

$$H_{\rm rot} = \frac{\omega_0}{2}\Big[\sin\theta\,\sigma_x + (\cos\theta - \varepsilon)\,\sigma_z\Big]. \tag{24}$$

This is a static field tilted to the angle θ' with

$$\tan\theta' = \frac{\sin\theta}{\cos\theta - \varepsilon}, \qquad \text{eigenvalues } \pm\frac{\omega_0}{2}\lambda, \qquad \lambda = \sqrt{1 - 2\varepsilon\cos\theta + \varepsilon^2}. \tag{25}$$

2.7 Non-adiabatic correction to the extracted phase

Start in $|+(\theta)\rangle$ and evolve for one period $T = 2\pi/\Omega$, so that $R(T) = e^{-i\pi\sigma_z} = -1$. In the eigenbasis $|\pm'\rangle$ of $H_{\rm rot}$,

$$|+(\theta)\rangle = \cos d\,|+'\rangle + \sin d\,|-'\rangle, \qquad d = \frac{\theta' - \theta}{2}. \tag{26}$$

Each component picks up $e^{\mp i\omega_0\lambda T/2} = e^{\mp i\pi\lambda/\varepsilon}$, so by Eq. (23)

$$\langle\psi(0)|\psi(T)\rangle = -\Big(\cos^2 d\, e^{-i\pi\lambda/\varepsilon} + \sin^2 d\, e^{+i\pi\lambda/\varepsilon}\Big). \tag{27}$$

The extracted geometric phase removes the adiabatic dynamical phase $-\frac{\omega_0}{2}T = -\pi/\varepsilon$:

$$\gamma_{\rm ext} = \arg\langle\psi(0)|\psi(T)\rangle + \frac{\pi}{\varepsilon}. \tag{28}$$

Now expand in ε. The tilt is $d = O(\varepsilon)$, so $\sin^2 d = O(\varepsilon^2)$ and the second term only enters at second order. For the first term,

$$\lambda = 1 - \varepsilon\cos\theta + \tfrac{1}{2}\varepsilon^2\sin^2\theta + O(\varepsilon^3) \quad\Longrightarrow\quad \frac{\pi\lambda}{\varepsilon} = \frac{\pi}{\varepsilon} - \pi\cos\theta + \frac{\pi}{2}\varepsilon\sin^2\theta + O(\varepsilon^2). \tag{29}$$

The divergent π/ε cancels and, modulo 2π,

$$\gamma_{\rm ext} = -\pi(1 - \cos\theta) - \frac{\pi}{2}\sin^2\theta\,\frac{\Omega}{\omega_0} + O(\varepsilon^2). \tag{30}$$

The linear correction is the $O(\Omega)$ shift of the rotating-frame energy, $\frac{\omega_0}{2}(\lambda - 1)$, which the adiabatic dynamical phase omits.

Aharonov–Anandan phase. If instead $\psi(0) = |+(\theta')\rangle$, the eigenstate of $H_{\rm rot}$, then χ only acquires a phase and $\psi = R\chi$ traces a cone of angle θ' exactly, at any Ω. The motion is cyclic, and subtracting the true dynamical phase $-\int_0^T\langle\psi|H|\psi\rangle dt$ leaves the geometric phase of that cone [4]:

$$\gamma_{\rm AA} = -\pi(1 - \cos\theta'). \tag{31}$$

The "non-adiabatic error" of Eq. (30) is therefore a matter of reference path: the state follows the tilted field of the rotating frame, and measured along that path the geometric phase is exactly half the solid angle.

2.8 Degenerate levels: the Wilczek–Zee connection

Let $\{D_a(\mathbf{R})\}$, $a = 1, \dots, d$, span a degenerate eigenspace at energy $E = 0$. Write $\psi = \sum_b c_b D_b$ and insert into the Schrödinger equation. The energy term vanishes, and projecting on $\langle D_a|$,

$$\dot{c}_a = -\sum_b \langle D_a|\dot{D}_b\rangle\, c_b = i \sum_b A^{ab} \cdot \dot{\mathbf{R}}\, c_b. \tag{32}$$

The $A_\mu = i\langle D_a|\partial_\mu D_b\rangle$ are now Hermitian $d \times d$ matrices that need not commute at different points, so the solution is a path-ordered exponential,

$$c(T) = U\, c(0), \qquad U = \mathcal{P} \exp\Big(i \oint A \cdot d\mathbf{R}\Big). \tag{33}$$

Under a change of basis $D \to DG(\mathbf{R})$ with G unitary, $A \to G^\dagger A G + iG^\dagger \partial G$ and $U \to G^\dagger(\mathbf{R}_0)\, U\, G(\mathbf{R}_0)$. Only the eigenvalues of U, not its matrix elements, are gauge invariant. The non-Abelian curvature is

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu - i[A_\mu, A_\nu], \tag{34}$$

which transforms covariantly, $F \to G^\dagger F G$. For a small counter-clockwise loop of area a in the (μ, ν) plane, $U = 1 + iF_{\mu\nu}a + O(a^{3/2})$.

2.9 Discrete Wilson loop

Let D_j be the 4×2 matrix of dark states at the jth point of the loop. Adiabatic transport keeps ψ in the dark subspace, so the components at the next point are obtained by projection:

$$c_{j+1} = D_{j+1}^\dagger D_j\, c_j = M_j^\dagger c_j, \qquad M_j = D_j^\dagger D_{j+1}. \tag{35}$$

Chaining the steps around the loop,

$$U = \big(M_0 M_1 \cdots M_{M-1}\big)^\dagger = W^\dagger. \tag{36}$$

Polar decomposition. Each link M_j is the overlap of two slightly rotated subspaces. Its singular values are cos of the principal angles, $1 - O(M^{-2})$, so the product of M links shrinks by $1 - O(M^{-1})$ and W is not unitary. Writing $M_j = V_j \Sigma_j W_j^\dagger$ (singular value decomposition) and replacing M_j by its unitary part $\mathrm{polar}(M_j) = V_j W_j^\dagger$ removes this:

$$U = \Big(\prod_j \mathrm{polar}(M_j)\Big)^\dagger. \tag{37}$$

The result is exactly unitary, coincides with discrete parallel transport (the closest unitary map between neighbouring subspaces), and converges as M^{-2}.

2.10 Tripod: dark states, connection and curvature

The tripod atom has an excited state $|0\rangle$ coupled to three ground states $|1\rangle, |2\rangle, |3\rangle$:

$$H = \sum_{j=1}^{3} \Omega_j |0\rangle\langle j| + \text{h.c.}, \tag{38}$$

$$\Omega_1 = \Omega_0 \sin\theta \cos\varphi, \qquad \Omega_2 = \Omega_0 \sin\theta \sin\varphi, \qquad \Omega_3 = \Omega_0 \cos\theta\, e^{iS}. \tag{39}$$

H couples $|0\rangle$ only to the bright state $|B\rangle \propto \sum_j \Omega_j^* |j\rangle$. The spectrum is $\{0, 0, \pm\Omega_0\}$; the two states orthogonal to $|0\rangle$ and $|B\rangle$ are dark. A smooth orthonormal dark basis is [6]

$$D_1 = e^{iS}\big(\sin\varphi|1\rangle - \cos\varphi|2\rangle\big), \tag{40}$$

$$D_2 = e^{iS}\cos\theta\big(\cos\varphi|1\rangle + \sin\varphi|2\rangle\big) - \sin\theta|3\rangle. \tag{41}$$

Differentiating and taking overlaps gives

$$A_\theta = 0, \qquad A_\varphi = \cos\theta\,\sigma_y, \qquad A_S = -\operatorname{diag}(1, \cos^2\theta). \tag{42}$$

The curvature components follow from the definition:

$$F_{\theta\varphi} = \partial_\theta A_\varphi = -\sin\theta\,\sigma_y, \tag{43}$$
$$F_{\theta S} = \partial_\theta A_S = \operatorname{diag}(0, \sin 2\theta), \tag{44}$$
$$F_{\varphi S} = -i[A_\varphi, A_S] = -\cos\theta\sin^2\theta\,\sigma_x. \tag{45}$$

For the last line, write $A_S = -\frac{1}{2}(1+\cos^2\theta)\mathbb{1} - \frac{1}{2}\sin^2\theta\,\sigma_z$; only the σ_z part contributes, and $[\sigma_y, \sigma_z] = 2i\sigma_x$. $F_{\varphi S}$ comes entirely from the commutator: it is a purely non-Abelian curvature. Its size is

$$\big\|[A_\varphi, A_S]\big\|_F = \sqrt{2}\cos\theta\sin^2\theta. \tag{46}$$

Why a phase must vary. If all Ω_j are real (S fixed), the dark states can be chosen real. Then $\langle D_a|\partial D_b\rangle$ is real and, by orthonormality, antisymmetric, so every A_μ is a multiple of the single matrix σ_y. All connections commute, the path ordering is irrelevant and every holonomy is a rotation about the same axis. Varying the phase S is necessary for non-Abelian behaviour.

Order of two loops. For two small loops a and b, $U_{a,b} \approx 1 + iF_{a,b}\,\mathrm{area}_{a,b}$, so

$$[U_a, U_b] \approx -[F_a, F_b]\,\mathrm{area}_a\,\mathrm{area}_b. \tag{47}$$

If both loops are scaled by λ the commutator grows as λ^4, and the infidelity between the two final states, which is quadratic in the commutator, as λ^8.

2.11 Adiabaticity and ramp shape

Leakage from the dark subspace to the bright states $|\pm\rangle$ at energy $\pm\Omega_0$ is driven by the non-adiabatic coupling $v\,\langle\pm|\partial D\rangle$, where v is the speed along the loop. First-order perturbation theory gives a leakage amplitude

$$a_\pm \sim \int v(t)\,\langle\pm|\partial D\rangle\,e^{\pm i\Omega_0 t}\,dt. \tag{48}$$

Integrating by parts, the boundary terms are v/Ω_0 at the start and end. For a linear ramp $v \sim 1/T$ is switched on and off abruptly, so $a \sim 1/(\Omega_0 T)$ and the infidelity is $\propto T^{-2}$, oscillating because the start and stop contributions interfere with relative phase $\Omega_0 T$. The ramp

$$s(u) = u - \frac{\sin 2\pi u}{2\pi}, \qquad u = t/T, \tag{49}$$

starts and stops with zero velocity, so the boundary terms vanish. A second integration by parts leaves terms in $\dot{v}/\Omega_0^2 \sim 1/(\Omega_0 T)^2$, and the infidelity falls as T^{-4}.

3 Numerical methods

3.1 Gauge-invariant geometric phases

- The Abelian Berry phase is computed from Eq. (15), with the eigenvectors u_j taken directly from a Hermitian eigensolver and no phase fixing.
- The Chern number is the sum of plaquette fluxes over a closed (θ, φ) grid, including the poles, where all states on a pole are the same up to phase.

- The non-Abelian holonomy is the product of polar-decomposed link matrices; the dark frames D_j are either the analytic basis above or an arbitrary orthonormal basis of the null space of H, obtained by singular value decomposition.
- Parallel transport, used as an independent reference, rotates each frame $D_{j+1} \to D_{j+1}G_{j+1}$ so that the link $D_j^\dagger D_{j+1}$ is Hermitian and positive, the discrete form of $\langle D_a|\partial D_b\rangle = 0$. The holonomy is then the overlap of the transported final frame with the initial one.
- The connection and curvature are also computed numerically, by central differences of the dark states, $\partial_\mu D \approx [D(x + h\hat{e}_\mu) - D(x - h\hat{e}_\mu)]/2h$, as an independent check of the analytic forms.

3.2 Fourth-order Magnus integrator

The propagator of $i\dot{\psi} = H(t)\psi$ over one step $[t_n, t_n + h]$ can be written as e^{Ω_M} with the Magnus series

$$\Omega_M = -i \int_{t_n}^{t_n+h} H(t_1)\, dt_1 - \frac{1}{2} \int_{t_n}^{t_n+h} dt_1 \int_{t_n}^{t_1} dt_2 \left[H(t_1), H(t_2)\right] + O(h^5). \tag{50}$$

Every truncation of the series is anti-Hermitian, so e^{Ω_M} is unitary at any order. The two integrals are evaluated with two-point Gauss–Legendre quadrature, at $t_{1,2} = t_n + \left(\frac{1}{2} \mp \frac{\sqrt{3}}{6}\right)h$, which is exact for polynomials of degree three. This gives [9]

$$\psi_{n+1} = e^{-iK}\psi_n, \qquad K = \frac{h}{2}\big(H_1 + H_2\big) - i\frac{\sqrt{3}}{12}h^2\big[H_2, H_1\big], \tag{51}$$

with $H_{1,2} = H(t_{1,2})$. K is Hermitian, so every step is exactly unitary: norm loss can never masquerade as leakage. The local error is $O(h^5)$ and the global error $O(h^4)$. The exponential e^{-iK} is formed from the eigendecomposition $K = V\,\mathrm{diag}(\kappa)V^\dagger$, done for all steps at once.

3.3 Spin dynamics

The spin is started in $|+(\theta)\rangle$ or $|+(\theta')\rangle$ and integrated over one period. The extracted phase is $\arg\langle\psi(0)|\psi(T)\rangle + \frac{\omega_0}{2}T$; the Aharonov–Anandan phase subtracts the dynamical phase $-\int\langle\psi|H|\psi\rangle dt$, evaluated by the trapezoid rule on the time grid of the integration. Both are compared with the closed forms from Eq. (23).

3.4 Four-level dynamics

The atom starts in $D_1(x_0)$ at the loop origin x_0. The full 4×4 Hamiltonian is integrated along the loop with the smooth ramp $s(u)$, and the final state is projected onto $D(x_0)$ and compared with the Wilson-loop prediction Uc_0 through the fidelity $F = |\langle Uc_0|c(T)\rangle|^2$. Leakage is the weight $1 - \|D^\dagger\psi(T)\|^2$ outside the dark subspace.

3.5 Loops, grids and step sizes

- *Cone loops:* M equally spaced azimuths $\varphi_j = 2\pi j/M$ at fixed θ, with M from 8 to 2000 for the convergence study and $M = 400$ for the main results.
- *Sphere grid for the Chern number:* θ_i equally spaced from 0 to π inclusive and φ_j periodic, 30×60 by default. At the poles all states are equal up to a phase, so the plaquettes that touch a pole are still well defined. The coarse-grid test uses 6×8 with a random phase on every state.
- *Tripod loops:* circles through the base point $x_0 = (\theta, \varphi, S) = (1.3, 0, 0)$, counter-clockwise in a coordinate plane, parametrised as $x(s) = x_0 + \rho\,(\cos 2\pi s - 1)\hat{e}_\mu + \rho \sin 2\pi s\, \hat{e}_\nu$ for $s \in [0, 1]$. Loop a lies in (θ, φ) with $\rho = 0.6$ and loop b in (φ, S) with $\rho = 1.5$; all loops keep $0 < \theta < \pi/2$, so every coupling amplitude stays positive. The Wilson loop uses $M = 2000$ frames.

- *Time evolution:* the loop parameter is $s = s(t/T)$ with the linear or smooth ramp, and the Magnus integrator uses $40\,T\Omega_0$ steps per loop, i.e. 40 steps per Rabi period $1/\Omega_0$.

4 Verification

The checks fall into six categories. Each compares a numerical object with an independent statement from Section 2.

1. Exact closed forms at finite resolution. Because Eq. (15) has the exact value $\gamma_M = -M \arg(1 - a + ae^{i\delta})$ on the cone, the discrete phase is tested to machine precision at every M, not only in the limit $M \to \infty$. A separate check confirms that the error against the continuum value scales as M^{-2} with the coefficient of Eq. (19).

2. Invariances that must hold exactly.

- *Gauge:* multiplying every u_j by a random phase must leave γ unchanged, and the Chern number must stay an integer on a coarse grid with random gauge.
- *Orientation:* reversing the loop must flip the sign of γ.
- *State:* the anti-aligned state must give the opposite phase, and at the poles γ must reach 0 and -2π.
- *Non-Abelian gauge covariance:* the holonomy eigenphases computed in the analytic basis and in an arbitrary null-space basis must agree, while the matrices themselves differ by a conjugation.

3. Properties of the model. H is Hermitian with eigenvalues $\pm\omega_0/2$; the spin state is a normalised eigenvector and single-valued in φ; the tripod spectrum is $\{0, 0, \pm\Omega_0\}$; the dark states are orthonormal and annihilated by H; and the analytic and null-space bases span the same subspace.

4. Analytic connection and curvature. A_μ and $F_{\mu\nu}$ obtained by central differences of the dark states are compared with the closed forms derived above, which checks the differentiation and the commutator term together. The small-loop identity $U = 1 + iF_{\mu\nu}\,a$ is checked in all three coordinate planes; it connects the curvature, a local derivative quantity, to the holonomy, a global product, and so tests the sign and normalisation of both.

5. Integrator. The Magnus step must be exact for a constant Hamiltonian; must reproduce the rotating-frame solution; and must be fourth order, which is tested by halving h and checking that the error falls by $2^4 = 16$. Then the extracted phase must match its closed form, approach the Berry phase linearly in ε with slope $-(\pi/2)\sin^2\theta$, and the Aharonov–Anandan phase must equal Eq. (31) at any speed.

6. Geometry against dynamics, and negative controls.

- The polar Wilson loop must be unitary and equal parallel transport; the raw product must approach it as M^{-2}.
- Slow four-level evolution must follow the holonomy, and the two loop orders must give the same final states in evolution as in the Wilson-loop prediction.
- *Negative controls:* with real couplings the holonomies must commute; the commutator must be nonzero on the demonstration loop; a link with orthogonal neighbours must be rejected rather than produce a meaningless phase; and the geometric phase must be reported as undefined when the evolved state has left the initial ray.

5 Results

5.1 Abelian Berry phase

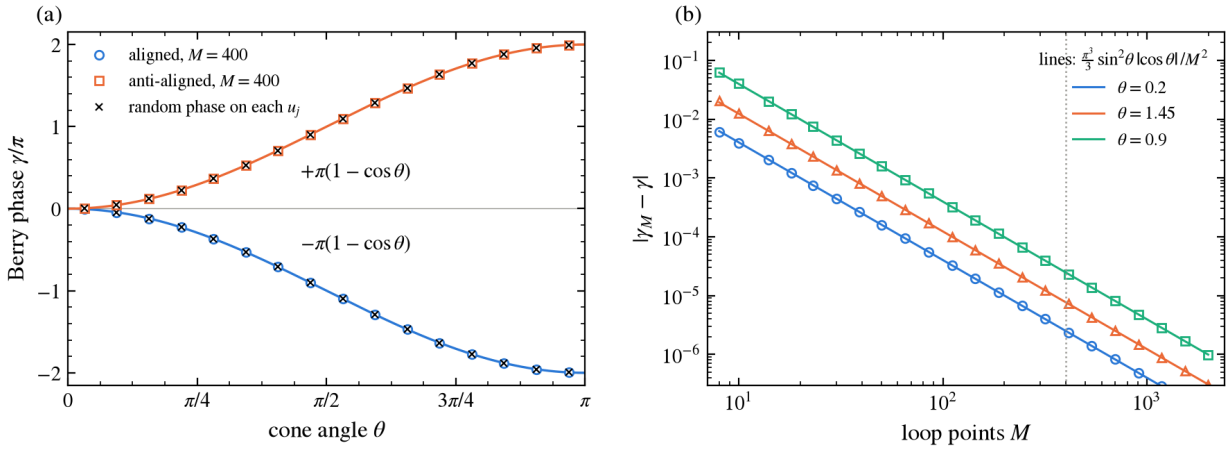


Figure 1: (a) Discrete Berry phase at $M = 400$ for aligned (circles) and anti-aligned (squares) states, with $\mp\pi(1-\cos\theta)$; crosses use a random phase on every u_j. (b) Error against M and the predicted leading term of Eq. (19) (lines, no fit).

Table 2: Discretisation error of the Berry phase.

θ	γ exact	$M = 20$	$M = 50$	$M = 400$	predicted, $M = 400$
0.3	-0.140315	2.15×10^{-3}	3.45×10^{-4}	5.389×10^{-6}	5.389×10^{-6}
0.9	-1.188747	9.90×10^{-3}	1.58×10^{-3}	2.464×10^{-5}	2.464×10^{-5}
1.5	-2.919365	1.84×10^{-3}	2.91×10^{-4}	4.547×10^{-6}	4.547×10^{-6}
2.2	-4.990423	-9.99×10^{-3}	-1.59×10^{-3}	-2.485×10^{-5}	-2.485×10^{-5}
2.9	-6.191948	-1.43×10^{-3}	-2.30×10^{-4}	-3.590×10^{-6}	-3.590×10^{-6}

The coefficient matches Eq. (19) to every printed digit and changes sign at $\theta = \pi/2$, which is why $\theta = 1.5$ is five times more accurate than $\theta = 0.9$. The Chern number is ∓ 1 to 12 digits even on a 6×8 grid with random gauge.

5.2 Non-adiabatic corrections

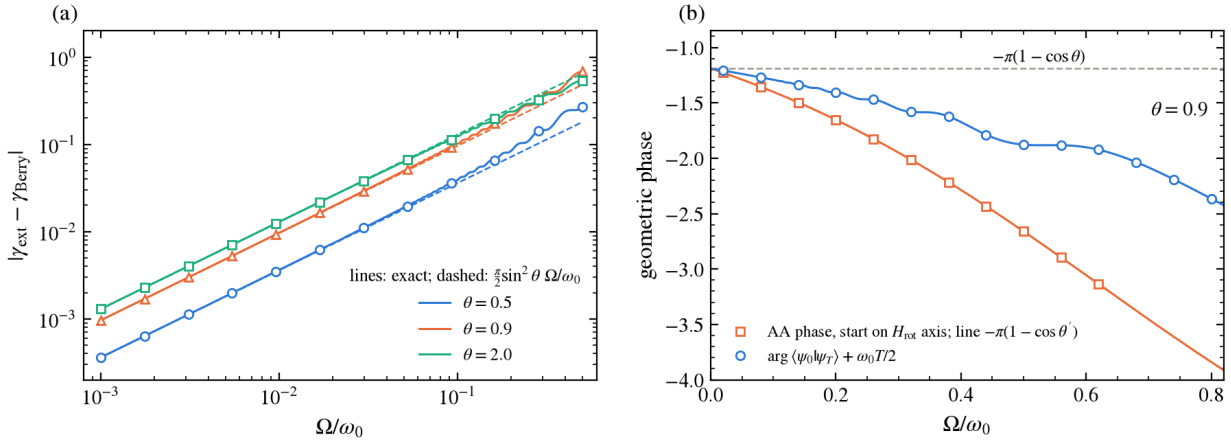


Figure 2: (a) $\gamma_{\rm ext} - \gamma$ against Ω/ω_0: Magnus integration (symbols), exact rotating-frame result (solid), leading term (dashed). (b) At $\theta = 0.9$: $\gamma_{\rm ext}$ and the Aharonov–Anandan phase with $-\pi(1-\cos\theta')$.

θ	log-log slope ($\varepsilon \le 0.007$)	$(\gamma_{\rm ext} - \gamma)/\varepsilon$ at $\varepsilon = 10^{-3}$	$-(\pi/2)\sin^2\theta$
0.5	1.0034	−0.36132	−0.36105
0.9	1.0028	−0.96434	−0.96384
2.0	0.9989	−1.29813	−1.29877

The remaining 0.05% is the $O(\varepsilon^2)$ term; integration agrees with the closed form to 1.4×10^{-7}. The steps above $\Omega/\omega_0 \approx 0.1$ are the $\sin^2 d$ admixture beating at $\lambda\omega_0$. The Aharonov–Anandan phase agrees with $-\pi(1 - \cos\theta')$ to 3.1×10^{-7} up to $\Omega/\omega_0 = 0.8$.

5.3 Wilson loop versus four-level evolution

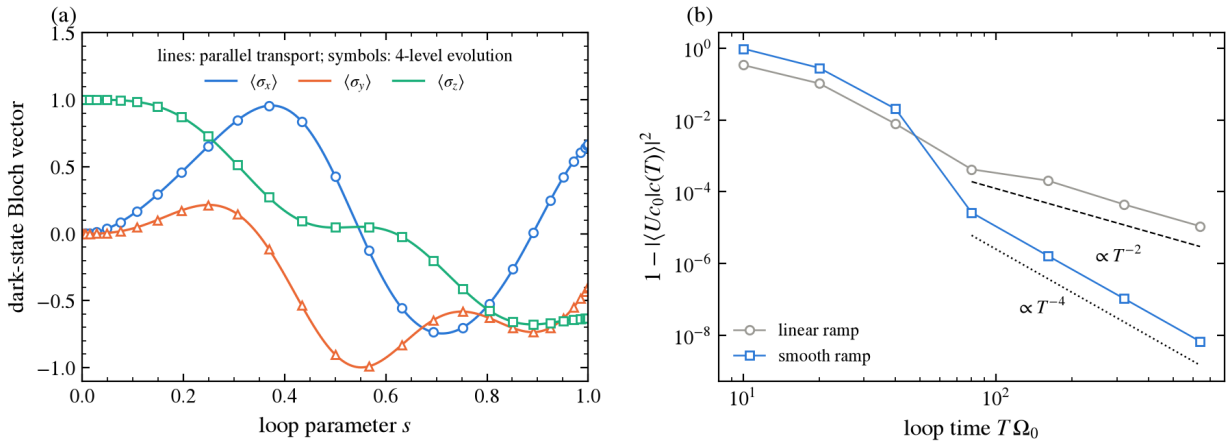


Figure 3: Loop b: radius-1.5 circle in (φ, S) at $\theta = 1.3$. (a) Bloch vector of the dark state in the local basis: parallel transport (lines, $M = 2000$) vs four-level evolution (symbols, $T\Omega_0 = 200$). (b) Infidelity for linear and smooth ramps with T^{-2} and T^{-4} guides.

Check	Result
Holonomy eigenphases, loop b	± 1.349018
$\|\,\mathrm{polar}(W)^\dagger - U\|$, $M = 250 \to 2000$	$5.5 \times 10^{-4} \to 8.5 \times 10^{-6}$ (M^{-2})
$1 -$ min singular value of raw W, $M = 2000$	9.2×10^{-3} (M^{-1})
$1 - F$ at $T\Omega_0 = 200$, smooth ramp	6.9×10^{-7}
Leakage out of dark subspace	1.2×10^{-10}
Fitted exponent of $1 - F$, linear / smooth	-1.80 / -3.97

At $T\Omega_0 = 640$ the smooth ramp reaches $1 - F = 7 \times 10^{-9}$ against 1.1×10^{-5} for the linear ramp. The holonomy contains no time scale: the same U predicts the final state for every T and both ramps.

5.4 Loop order: non-Abelian holonomy

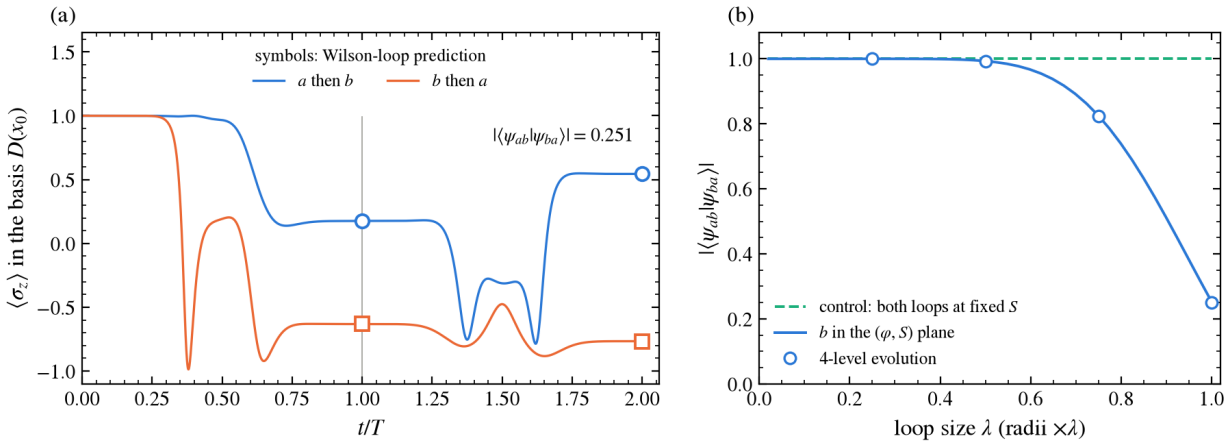


Figure 4: Loop a in (θ, φ) (radius 0.6), loop b in (φ, S) (radius 1.5), both from $x_0 = (1.3, 0, 0)$. (a) $\langle\sigma_z\rangle$ during the four-level evolution, a then b (blue) and b then a (orange); symbols are Wilson-loop predictions. (b) Overlap of the two final states when both radii are scaled by λ; dashed: Abelian control.

	a then b	b then a
$1 - F$ vs Wilson prediction	2.9×10^{-8}	2.7×10^{-6}
Overlap $\lvert\langle\psi_{ab}\rvert\psi_{ba}\rangle\rvert$, Wilson / evolution	0.2523 / 0.2508	
$\lVert[U_a, U_b]\rVert_F$	1.770	
Control (S fixed): $\lVert[U_a, U_{b'}]\rVert_F$, overlap	7.4×10^{-14}, 1.000000000000	

From $\lambda = 0.5$ to 0.75 the infidelity $1 - |\langle\psi_{ab}|\psi_{ba}\rangle|$ grows by 25.7, against $1.5^8 = 25.6$ from the λ^8 law. Two loops traversed in opposite order, with no energy difference anywhere, leave the atom in states that are 75% orthogonal, while the same experiment with real couplings gives overlap 1 to twelve digits.

6 Discussion

Gauge invariance needs a closed loop. Under $u \to e^{i\alpha(\mathbf{R})}u$, γ changes by $\alpha(\text{end}) - \alpha(\text{start})$; on a closed loop this is $2\pi n$. In Eq. (15) the endpoint phases of an open path survive.

Why the phase is geometric. $\gamma = \oint A \cdot d\mathbf{R}$ contains no time, so any reparametrisation leaves it unchanged; the dynamical phase scales with T. By Stokes, γ is the curvature flux through the loop; for the spin it is a monopole of charge $\frac{1}{2}$, and the surface ambiguity $4\pi \times \frac{1}{2} = 2\pi$ is invisible in $e^{i\gamma}$.

Relation to the SSH Zak phase. The Zak phase is the same calculation with crystal momentum as the loop parameter; the SSH project uses the same discrete formula, where chiral symmetry quantises it to 0 or π.

Limitations. The tripod is ideal: no spontaneous emission from $|0\rangle$, no detuning, perfect degeneracy. Detuning would lift the dark-state degeneracy and add a dynamical phase inside the subspace; this is a natural extension. The small-loop test in the (φ, S) plane converges only as $O(r)$ because of path ordering, so it constrains $F_{\varphi S}$ less tightly than the other components.

7 Conclusion

Abelian and non-Abelian geometric phases were computed by formulas immune to eigensolver gauge freedom and checked against closed forms, including an exact finite-M expression, and against independent time evolution. The non-adiabatic correction was derived and matched, and recast via the Aharonov–Anandan phase as exact geometry of the actual path. For the tripod, the polar Wilson loop, analytic curvature and four-level dynamics agree, and the order-dependence of two loops demonstrates a non-Abelian U(2) gauge field, absent when all couplings are real.

References

[1] M. V. Berry, *Quantal phase factors accompanying adiabatic changes*, Proc. R. Soc. Lond. A **392**, 45 (1984).

[2] F. Wilczek, A. Zee, *Appearance of gauge structure in simple dynamical systems*, Phys. Rev. Lett. **52**, 2111 (1984).

[3] B. Simon, *Holonomy, the quantum adiabatic theorem, and Berry's phase*, Phys. Rev. Lett. **51**, 2167 (1983).

[4] Y. Aharonov, J. Anandan, *Phase change during a cyclic quantum evolution*, Phys. Rev. Lett. **58**, 1593 (1987).

[5] D. Xiao, M.-C. Chang, Q. Niu, *Berry phase effects on electronic properties*, Rev. Mod. Phys. **82**, 1959 (2010).

[6] J. Ruseckas, G. Juzeliūnas, P. Öhberg, M. Fleischhauer, *Non-Abelian gauge potentials for ultracold atoms with degenerate dark states*, Phys. Rev. Lett. **95**, 010404 (2005).

[7] J. Dalibard, F. Gerbier, G. Juzeliūnas, P. Öhberg, *Colloquium: Artificial gauge potentials for neutral atoms*, Rev. Mod. Phys. **83**, 1523 (2011).

[8] T. Fukui, Y. Hatsugai, H. Suzuki, *Chern numbers in discretized Brillouin zone*, J. Phys. Soc. Jpn. **74**, 1674 (2005).

[9] S. Blanes, F. Casas, J. A. Oteo, J. Ros, *The Magnus expansion and some of its applications*, Phys. Rep. **470**, 151 (2009).